# Bank Customer Segmentation Using Clustering

## Introducing the Problem

In today's digital banking world, understanding customers is more important than ever. Banks collect huge amounts of data from their clients such as age, job, education, and account balance but it's not always clear how to use that information effectively. My goal in this project is to use clustering to uncover meaningful customer groups within a bank's marketing data.

By segmenting customers based on their demographics and financial behavior, I hope to identify patterns such as loyal savers, young professionals, or potential investors. Recognizing these groups can help financial institutions design smarter marketing strategies, personalize products, and improve customer relationships.

I chose this problem because it reflects a real challenge in data-driven industries: making sense of large, complex datasets to better serve people. Through this project, I want to explore how unsupervised learning, specifically clustering can turn raw marketing data into practical business insights.

## What Is Clustering and How Does It Work?

Clustering is an unsupervised machine learning technique that groups data points based on their similarity. Unlike classification, clustering doesn't rely on predefined labels; it looks for hidden patterns in the data and organizes items that behave alike into the same group.

In this project, I'm using clustering to uncover natural customer segments within a bank's marketing data. Two popular algorithms I'll explore are K-Means and Agglomerative (Hierarchical) Clustering.

K-Means works by dividing the data into K clusters. It starts by choosing random "centroids" and then repeatedly adjusts them so that each customer belongs to the group whose centroid is closest in terms of distance, usually measured with Euclidean distance. The goal is to minimize variation within clusters and maximize the differences between them.

Agglomerative Clustering, on the other hand, begins with each customer as its own cluster and then merges the most similar clusters step by step until only a few large groups remain. It's visualized with a dendrogram, which shows how clusters merge and helps decide the ideal number of groups.

By comparing these two methods, I hope to find the most meaningful way to represent customer patterns and better understand how people's financial behaviors naturally group together.

**Introduce the Data**

For this project, I am using the **UCI Bank Marketing Dataset**, which I accessed through Kaggle. The dataset contains information collected from a series of marketing campaigns conducted by a Portuguese bank. Each row represents a client who was contacted by phone during a campaign, and the goal was to determine whether they would subscribe to a term deposit.

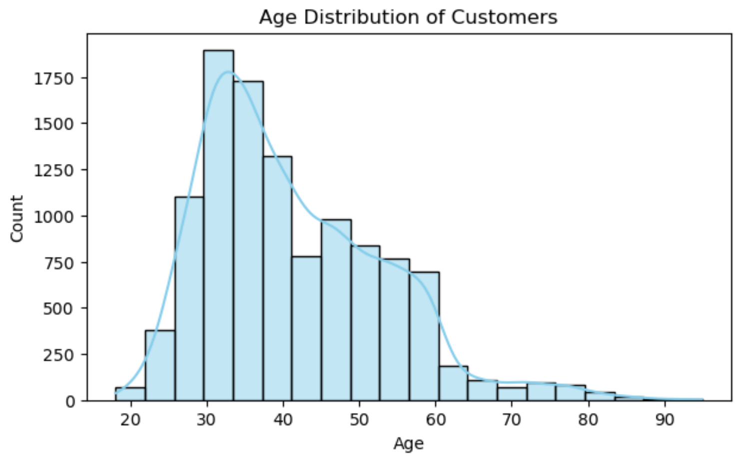
The dataset includes over 45,000 records and 17 features describing both personal and campaign-related details. Some of the main features are:

- age: The client's age in years
- job: Type of job (admin, technician, management, blue-collar, etc.)
- marital: Marital status (single, married, divorced)
- education: Level of education (primary, secondary, tertiary)
- balance: Average yearly account balance
- housing / loan: Whether the client has a housing loan or personal loan
- contact, day, month, duration, campaign: Information about the contact communication and marketing campaign
- pdays / previous / poutcome: Details about previous campaign interactions
- deposit: Indicates if the client subscribed to a term deposit

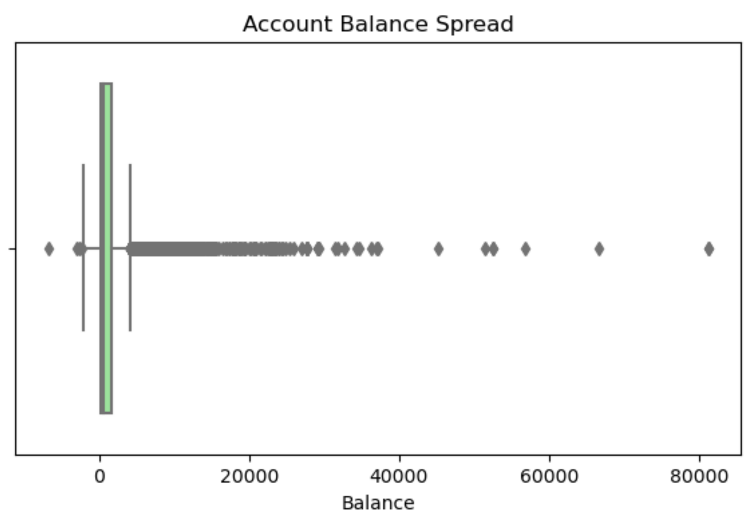
This dataset is valuable because it combines demographic, financial, and marketing interaction data, giving a complete view of customer behavior. It provides a strong foundation for clustering, as these features can help reveal patterns among clients who share similar financial profiles or campaign responses.

**Data Understanding and Visualization**

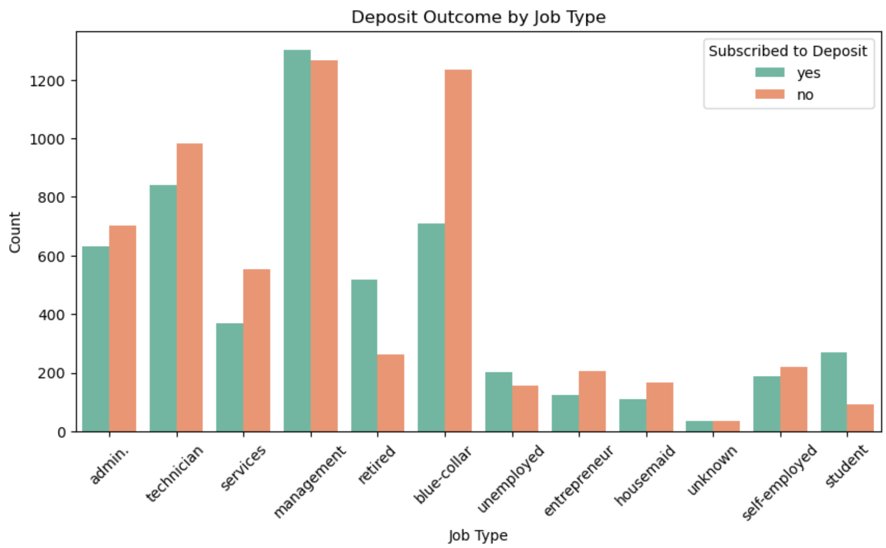
Before clustering, I explored the dataset to understand customer characteristics and behaviors that might influence their response to marketing campaigns. The dataset contains 11,162 records and 17 columns, with no missing values. This confirmed that the data is clean and ready for deeper analysis.



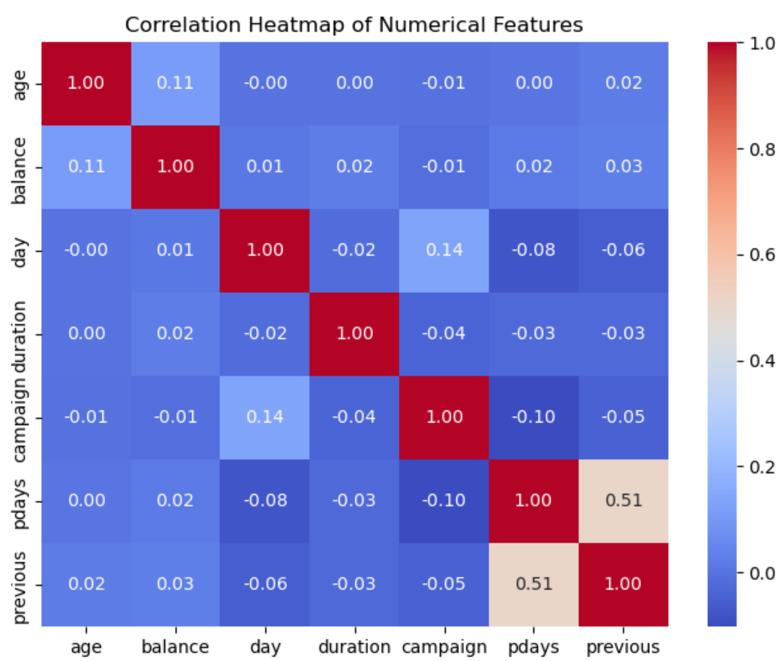

Most customers are between 30 and 50 years old, with the highest concentration around the early 30s. There are fewer customers over 60, suggesting that the bank's marketing mainly targets middle-aged individuals who are more likely to invest in financial products. This helps guide modeling because age can be an important factor in identifying customer segments like "young professionals" or "retirees."



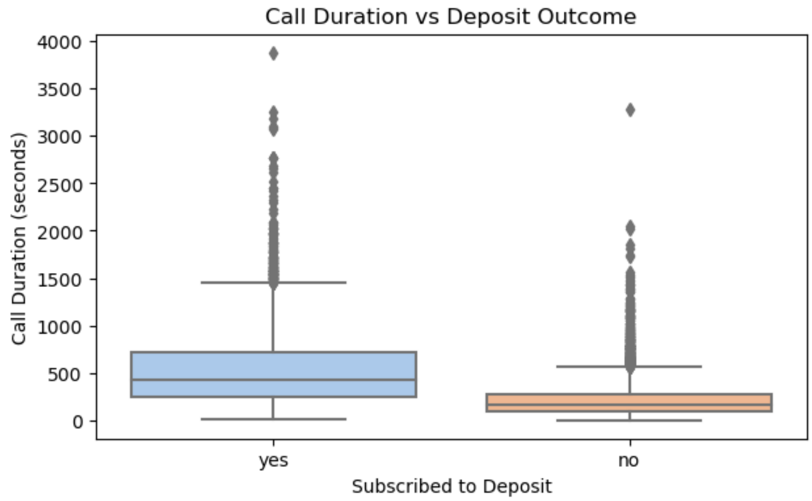

The balance distribution shows many customers with low or moderate balances, while a few have extremely high values (some exceeding 80,000). This indicates strong variation in financial status across the customer base. These differences are important for clustering since balance can help separate groups such as "high-value customers" and "low-income customers."




Customers in management and technician roles have higher subscription rates for term deposits, while those in blue-collar or services roles show lower engagement. This suggests that occupation and income stability may influence customers' willingness to invest. Including job type in clustering can help the model form segments based on employment and lifestyle patterns.




Most numeric variables are weakly correlated, except for *pdays* and *previous*, which share a moderate positive relationship. This makes sense because both track previous campaign contacts. The low overall correlation indicates that each feature may contribute unique information to the clustering process, helping create distinct customer profiles.




Customers who subscribed to a deposit generally had longer call durations compared to those who didn't. This likely means that more interested or financially stable customers spend more time discussing offers with bank representatives. This feature could be used to differentiate customers by engagement level or interest in financial products.

How This Step Relates to Modeling

Exploring the data helped me identify which features carry meaningful behavioral signals. Based on these insights, I plan to include demographic and financial variables such as age, balance, job type, and duration in my clustering model. Understanding these distributions ensures that the clusters represent true customer segments instead of random groupings influenced by outliers or redundant variables.

**Preprocessing the Data**

Before moving on to clustering, I made sure the dataset was clean, consistent, and ready to use. Since clustering algorithms like K-Means rely on measuring distances between data points, it's important that all features are on a similar scale and in numeric form.

I began by checking for any missing or "unknown" values in columns such as job, education, and outcome. Instead of deleting rows, I filled these entries with the most common category so that I could keep all records while maintaining data quality.

Next, I converted the categorical variables into numbers using label encoding. This allowed the algorithm to understand text-based information such as marital status or job type by assigning each unique label a numerical value. After encoding, I focused on the features that best describe customer behavior and financial status: age, job, marital status, education, balance, housing, loan, and duration.

Finally, I standardized all numerical columns using the StandardScaler method. This adjusted each variable so that it had an average value of zero and a consistent spread, making sure no single feature (like balance) dominated the others. After scaling, the dataset looked clean, balanced, and ready for modeling. This preparation step ensures that the clustering process will focus on real customer patterns rather than uneven data ranges or formats.

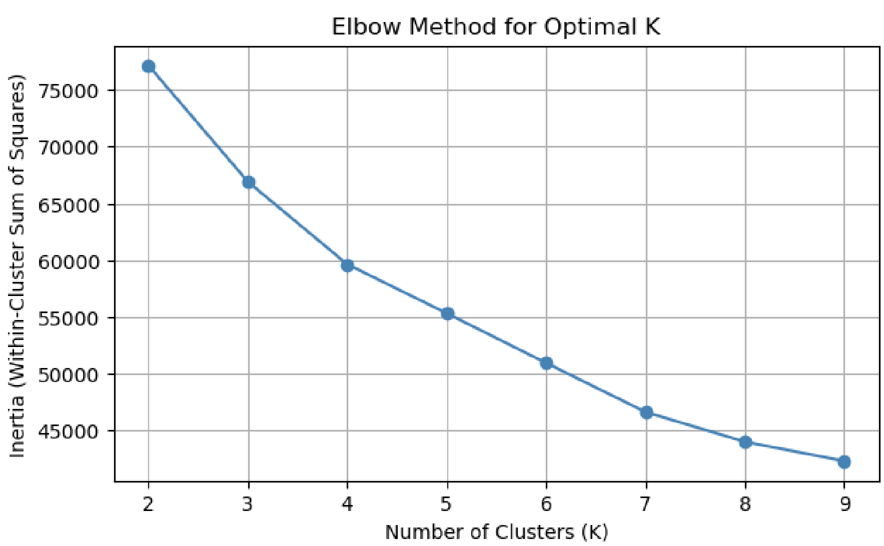
**Modeling (Clustering)**

After preprocessing the data, I moved on to the modeling stage to identify meaningful customer groups. I began with the **K-Means clustering algorithm**, which groups customers based on similarities across their financial and demographic features.

Elbow Method for Optimal K

The **Elbow Method** was used to determine the ideal number of clusters. As shown in **Figure 1**, the curve begins to level off at **k = 3**, meaning that three clusters provide a good balance between model simplicity and accuracy. Beyond this point, adding more clusters results in only small improvements in within-cluster variation.

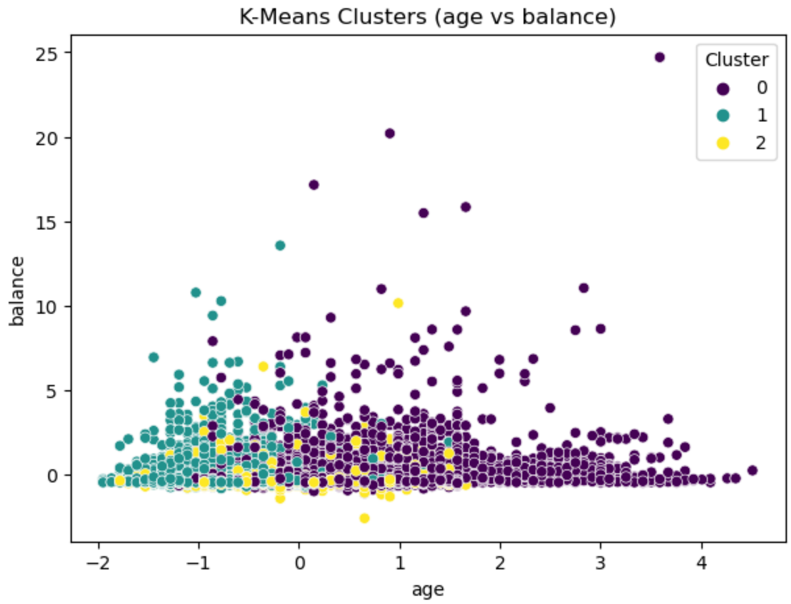
*(Figure 1: Elbow Method for Optimal K)*




K-Means Clusters (Age vs Balance)

Using **k = 3**, the K-Means algorithm divided the data into three clear clusters. In **Figure 2**, each color represents a separate customer group. Cluster 0 mostly contains middle-aged or older customers with higher account balances, Cluster 1 represents younger customers with moderate balances, and Cluster 2 includes younger clients with lower balances. These divisions show how age and financial standing influence customer behavior.

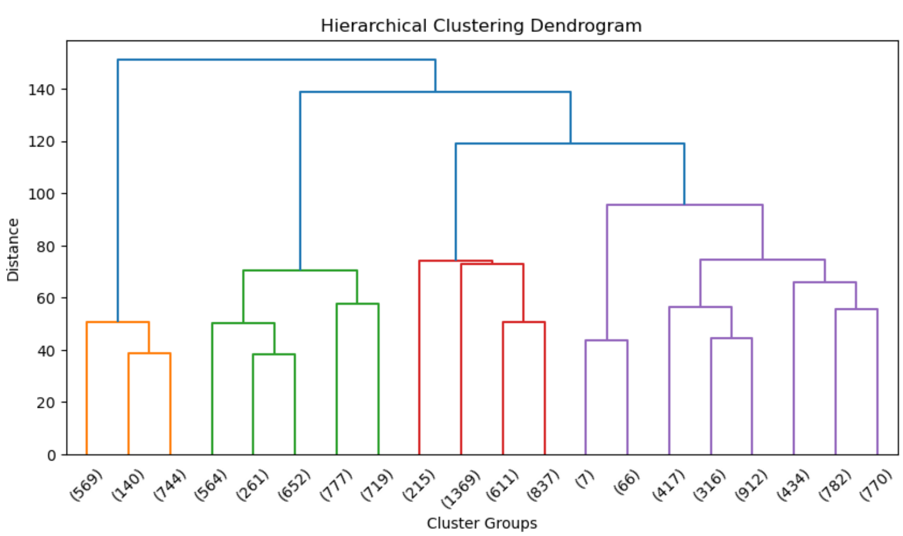
*(Figure 2: K-Means Clusters – Age vs Balance)*




Hierarchical Clustering Dendrogram

Next, I applied **Agglomerative (Hierarchical) Clustering** to visualize how clusters merge step by step. The **dendrogram** in **Figure 3** shows the merging distance between clusters, with a noticeable cutoff forming around three main groups—consistent with the Elbow Method. This confirms that customers naturally fall into about three major segments based on their characteristics.

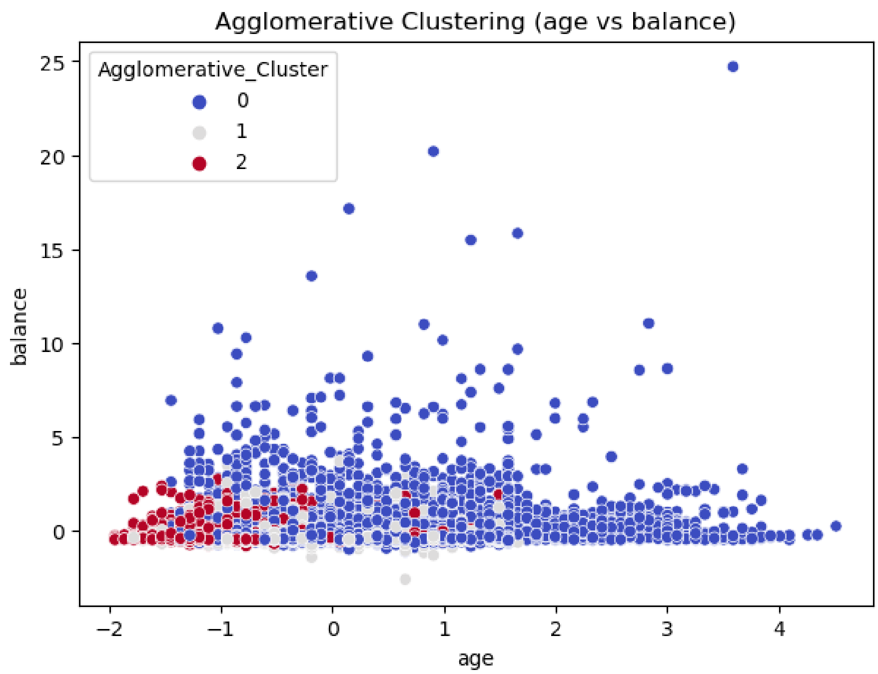
*(Figure 3: Hierarchical Clustering Dendrogram)*



Agglomerative Clustering (Age vs Balance)

The **Agglomerative Clustering** scatter plot in **Figure 4** shows results like the K-Means model. Cluster 0 represents the largest customer group with varied ages and moderate balances, Cluster 1 includes customers with slightly lower balances, and Cluster 2 highlights younger clients with small balances or debts. The consistency between both clustering techniques adds confidence that the results are stable and meaningful.

*(Figure 4: Agglomerative Clustering – Age vs Balance)*




Summary of Findings

Both K-Means and Agglomerative Clustering revealed three primary customer groups that share similar behaviors and financial profiles. These clusters can help banks tailor their marketing efforts, for example, offering investment opportunities to high-balance clients while creating savings-oriented plans for younger customers with lower balances.

**Storytelling (Clustering Analysis)**

The clustering analysis uncovered meaningful patterns in customer behavior and financial characteristics. By applying both K-Means and Agglomerative Clustering, I was able to divide the bank's clients into <u>three clear segments</u>, each representing a distinct financial profile and engagement level with the bank.

<u>The Established Investors</u> form the first and most financially stable group. These customers are typically older, have higher account balances, and tend to spend more time during

campaign calls. Their strong financial position and willingness to invest make them the bank's most profitable and loyal segment. They are ideal candidates for premium savings accounts, investment portfolios, and long-term deposit plans.

The Mid-Level Professionals make up the second group. These customers are usually in their 30s and 40s, employed in steady jobs, and maintain moderate account balances. They show consistent engagement but are often cautious about high-risk investments. This group has strong potential for future financial growth, and with proper guidance or financial education, they could transition into higher-value investors over time.

The Young Budget Savers represent the third segment. These customers are generally younger, early in their careers, and have lower balances and shorter campaign interactions. They are not yet heavily involved in financial products, but they hold long-term potential as their income and financial awareness grow. Building loyalty and offering starter savings or credit programs could help this group mature into future investors.

These results directly answered my initial question about how to segment customers based on their demographic and financial data to improve marketing strategies. The analysis revealed that factors such as age, account balance, and housing or loan status strongly influence investment behavior and responsiveness to campaigns.

Overall, this project showed that clustering is a powerful way to uncover natural patterns within customer data. By identifying these segments, banks can design more personalized campaigns, allocate resources more efficiently, and strengthen long-term relationships with each type of client.

**Impact Section**

This project gave me a clearer understanding of how data-driven methods like clustering can make banking more personalized and efficient. By identifying patterns in customer behavior, banks can design better marketing strategies, offer products that truly fit each customer's needs, and strengthen long-term relationships. For example, promoting investment plans to financially established clients or offering savings programs to younger customers can make campaigns more targeted and meaningful.

At the same time, it's important to think about the ethical side of using customer data. Privacy should always come first. Any data used for analysis needs to be kept secure and anonymous to protect clients' personal information. Another key point is avoiding bias if the data reflects existing inequalities or certain demographic patterns, the model could unintentionally favor one group over another.

There are also broader social effects to consider. When used thoughtfully, clustering can help promote financial inclusion by helping banks reach people who might otherwise be overlooked. But if used carelessly, it could also lead to unfair targeting or exclusion.

Overall, this project showed me that while clustering can bring valuable insights and improve decision-making, it should always be paired with fairness, transparency, and respect for privacy. Used responsibly, these insights can help banks serve customers better and build stronger, more ethical business practices.

**References**

https://www.kaggle.com/datasets/janiobachmann/bank-marketing-dataset